ATOMIC-LEVEL MATERIALS SIMULATION

Module Code: MES306TC

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Brief Introduction to Quantum Mechanics I

I. What is the origin of this quantum mechanics?

II. What is wave function?

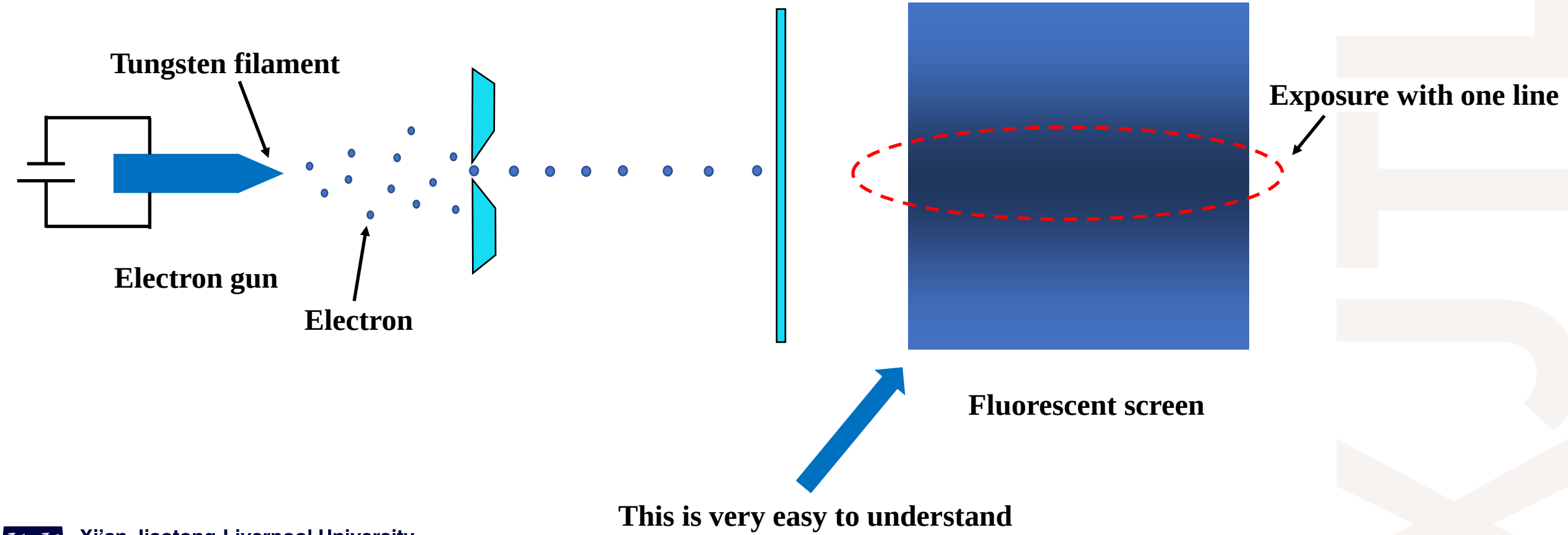
III. What is the representation transformation?

IV. What is the expectation value of operator?

V. What is density operator?

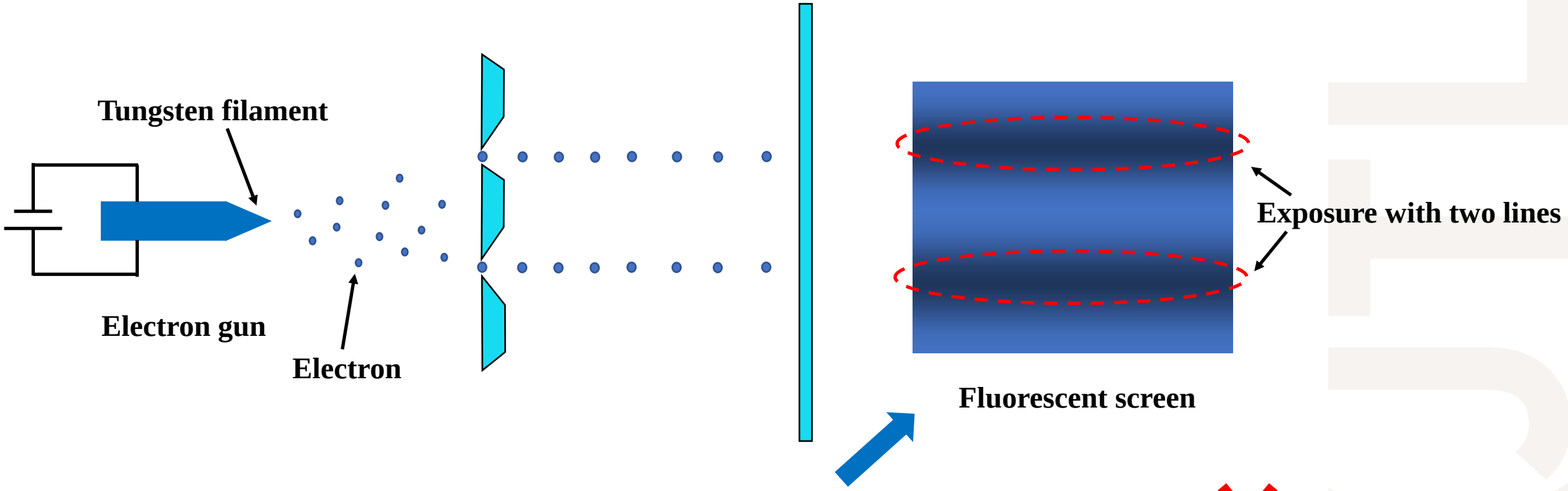
I. What is the origin of this quantum mechanics?

Experiment on the Electron Double-Slit Interference



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Experiment on the Electron Double-Slit Interference

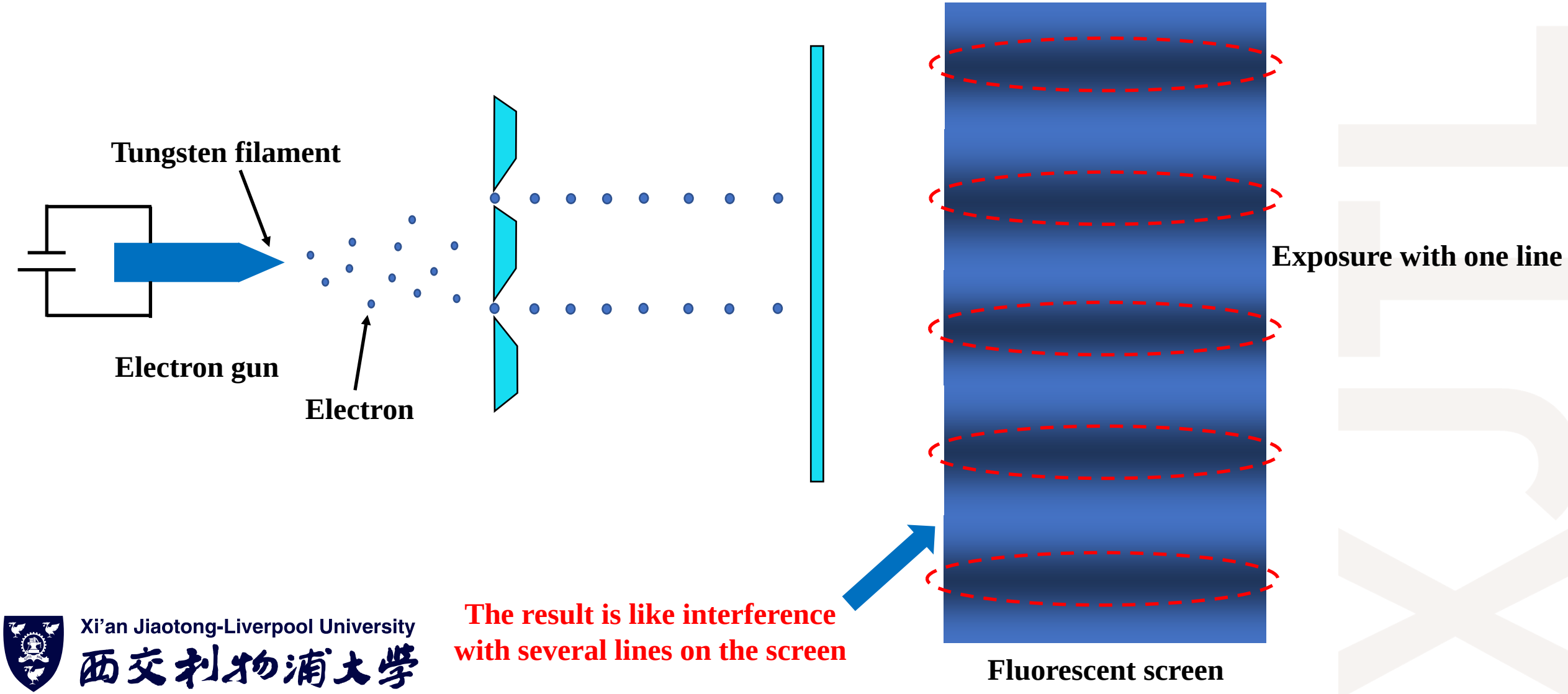


The result is supposed to be like two lines on the screen ✗

Actually, the result is completely different from this!!!

I. What is the origin of this quantum mechanics?

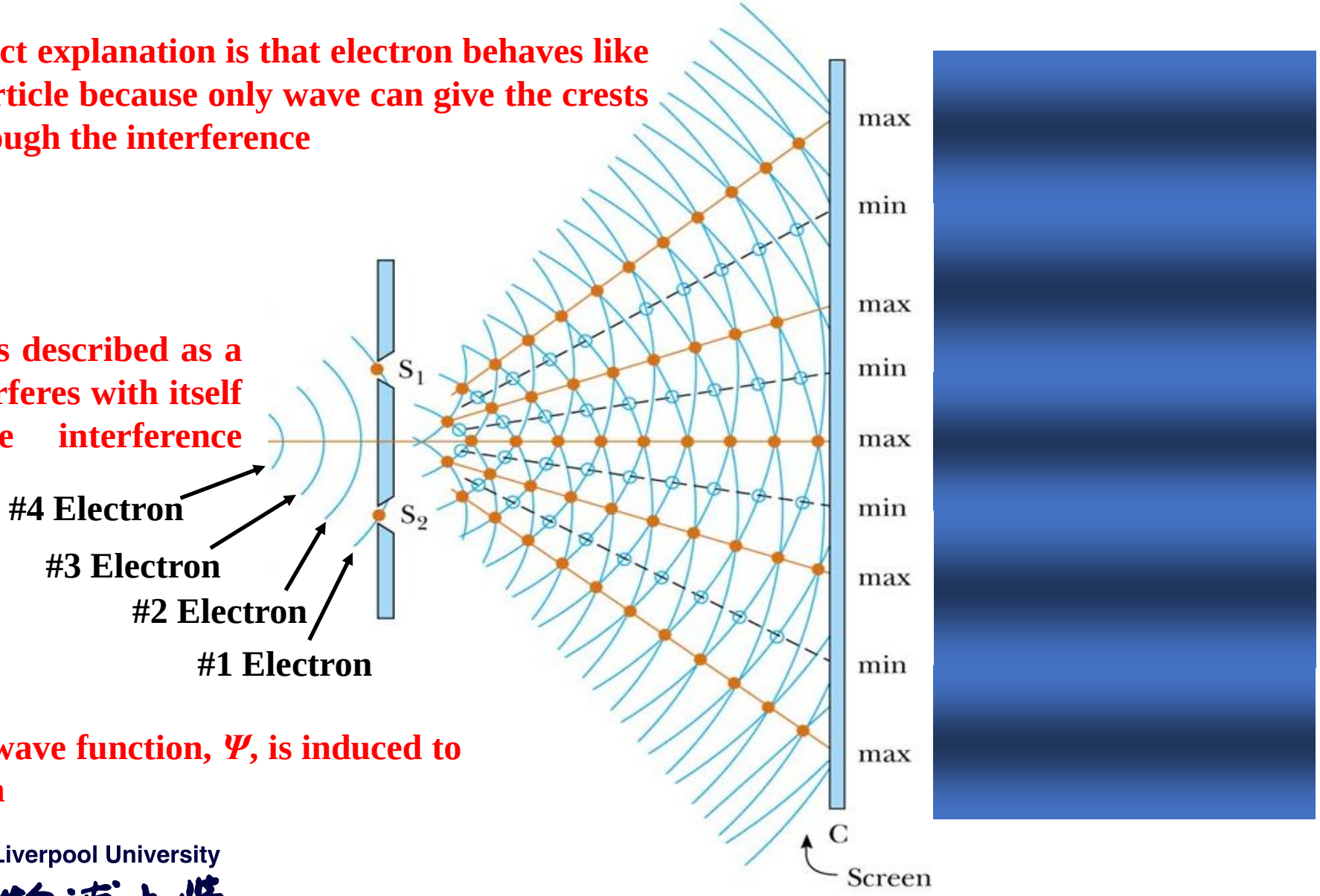
Experiment on the Electron Double-Slit Interference



I. What is the origin of this quantum mechanics?

♦ The only correct explanation is that electron behaves like wave; rather particle because only wave can give the crests and troughs through the interference

♦ One electron is described as a wave and it interferes with itself to induce the interference pattern



♦ The notion of wave function, Ψ, is induced to describe electron

II. What is wave function?

Each electron interferes with itself

+

Each electron reaches the screen as one integrity

→ Electron cannot be smeared out

The wavefunction, Ψ, does not imply that a physical electron is physically split or smeared out like a water wave, but rather that the probability density of finding the electron at a specific location in space possesses wave characteristics

The wavefunction, Ψ

Key notes

- It is a function of position and time, $\Psi(r,t)$.
- It does not mean something vibrating in reality
- It is just the probability amplitude
- Its modulus squared, $|\Psi|^2$, represents the probability density
  - The probability to find particle in position r and instant t

II. What is wave function?

Not only electron but all particles (photon, phonon, magnon *etc*.) can be described by wavefunction

The wavefunction can be expanded in terms of any complete set of orthonormal basis states

$$\Psi = \sum_i c_i |b_i\rangle$$

$$\langle b_i | b_i \rangle = 1$$
$$\langle b_i | b_j \rangle = 0$$

In a Hilbert space, a set of basis states $|b_i\rangle$ is said to be complete if any arbitrary state Ψ can be represented as a linear combination of the basis vectors

The basis state is expressed as a complex vector matrix $|b_i\rangle = \begin{pmatrix} e^{i\theta_1^i} \\ \vdots \\ e^{i\theta_n^i} \end{pmatrix}$

$\langle b_i|$ is the conjugate transpose of $|b_i\rangle$, $\langle b_i| = (e^{-i\theta_1^i} \quad \dots \quad e^{-i\theta_n^i})$

The wavefunction can be expanded in the form of matrix $\Psi = \begin{pmatrix} c_1 e^{i\theta_1^{\vdots}} & c_2 e^{i\theta_1^{\vdots}} & \cdots & c_n e^{i\theta_1^{\vdots}} \\ \vdots & \vdots & & \vdots \\ c_1 e^{i\theta_n^{\vdots}} & c_2 e^{i\theta_n^{\vdots}} & & c_n e^{i\theta_n^{\vdots}} \end{pmatrix}$

COMPARISON

Classical Mechanics

1. Macroscopic objects (celestial bodies, cars, fluids)
2. Determinism
3. Continuity
4. Only particle property
5. Certainity
6. Locality

Quantum Mechanics

1. Microscopic objects (atoms, electrons, photons, energy bands inside chips)
2. Probability
3. Quantization
4. Wave-Particle Duality
5. Uncertainty Principle
6. Non-locality (Quantum Entanglement)

III. What is the Representation transformation?

Wavefunction can be expanded in terms of any arbitrary basis

Presentation of the wavefunction $|b_i\rangle$ basis $\Psi = \sum_i c_i |b_i\rangle$

Presentation of the wavefunction $|e_i\rangle$ basis $\Psi = \sum_j d_j |e_j\rangle$

$$\sum_i c_i |b_i\rangle = \sum_j d_j |e_j\rangle$$

$$\sum_i \langle b_i | c_i | b_i \rangle = \sum_j \langle b_i | d_j | e_j \rangle$$

$$\sum_i c_i \langle b_i | b_i \rangle = \sum_j d_j \langle b_i | e_j \rangle$$

Expression of wavefunction from one basis to another different one

$$c_i = \sum_j d_j \langle b_i | e_j \rangle$$

Representation transformation

III. What is the Representation transformation?

$$c_i = \sum_j d_j \langle b_i | e_j \rangle$$



Matrix Form

$$\begin{pmatrix} \boldsymbol{c_1} \\ \boldsymbol{c_2} \\ \vdots \end{pmatrix} = \begin{pmatrix} \langle \boldsymbol{b_1} | \boldsymbol{e_1} \rangle & \langle \boldsymbol{b_1} | \boldsymbol{e_2} \rangle & \cdots \\ \langle \boldsymbol{b_2} | \boldsymbol{e_1} \rangle & \langle \boldsymbol{b_2} | \boldsymbol{e_2} \rangle & \cdots \\ \vdots & \vdots & \cdots \end{pmatrix} \begin{pmatrix} \boldsymbol{d_1} \\ \boldsymbol{d_2} \\ \vdots \end{pmatrix}$$

III. What is the Representation transformation?

Why transformation?

A quantum state Ψ is a physical reality in Hilbert space, independent of any coordinates
Choosing a basis is merely choosing a "perspective" to observe it

♦ Some physical problems are easy to solve in Position space (e.g., potential wells V(x))

♦ Other problems are much simpler in Momentum space (e.g., free particles, scattering problems)

♦ Representation Transformation is the mathematical bridge that allows us to switch back and forth to find the easiest path to a solution

♦ Analogy: It is exactly like switching from Cartesian coordinates (x, y, z) to Spherical coordinates (r, θ, φ) in classical mechanics to solve a hydrogen atom problem

III. What is the Representation transformation?

Continuous Representation Transformation

In maths, many quantum bases are continuous (like position $|x\rangle$ or momentum $|p\rangle$)
The summation Σ naturally becomes an integral ∫

Example: $\Psi(x) = \int \langle x|p\rangle \phi(p) dp$

Here, $\Psi(x) = \langle x|\Psi\rangle$ **is the wavefunction in position representation, and** $\phi(p) = \langle p|\Psi\rangle$ **is the wavefunction in momentum representation**

III. What is the Representation transformation?

Example: Position Space ⟷ **Momentum Space**

In quantum mechanics, a momentum eigenstate $|p\rangle$ expressed in position space is a plane wave

$$\langle x|p\rangle = \frac{1}{\sqrt{2\pi\hbar}} e^{ipx/\hbar}$$

Using the continuous completeness relation, we can find the momentum wavefunction $\phi(p)$ from the position wavefunction $\Psi(x)$

$$\phi(p) = \langle p|\Psi\rangle = \int \langle p|x\rangle\langle x|\Psi\rangle dx = \frac{1}{\sqrt{2\pi\hbar}} \int e^{-ipx/\hbar}\Psi(x)dx$$

The representation transformation between position space and momentum space is mathematically equivalent to a Fourier Transform

IV. What is the expectation value of operator?

The expectation value of any operator ($\hat{A}$) is the measurement value of this physical observable in reality

Average value

For example, velocity of any particle, position of any particle, energy of any particle, *etc*.

In Quantum Mechanics, the operator ($\hat{A}$) is the described as a matrix as well as the wavefunction

The expectation value of any operator ($\hat{A}$) is calculated as $\frac{\langle \Psi | \hat{A} | \Psi \rangle}{\langle \Psi | \Psi \rangle}$ **Normalization factor**

To make sure the value is normalized to 1

If the wavefunction is the eigenstate of the system, then, the expectation value of any operator ($\hat{A}$) is calculated as $\langle \phi | \hat{A} | \phi \rangle$ because eigenstate is normalized to 1 ($\langle \phi | \phi \rangle = 1$)

IV. What is the expectation value of operator?

Taking the momentum operator as an example $\hat{\boldsymbol{p}} = -\boldsymbol{i}\hbar\boldsymbol{\nabla} = -\boldsymbol{i}\hbar\left(\hat{\boldsymbol{\imath}}\dfrac{\boldsymbol{\partial}}{\boldsymbol{\partial x}} + \hat{\boldsymbol{\jmath}}\dfrac{\boldsymbol{\partial}}{\boldsymbol{\partial y}} + \hat{\boldsymbol{k}}\dfrac{\boldsymbol{\partial}}{\boldsymbol{\partial z}}\right)$

The matrix form of the momentum operator $\hat{\boldsymbol{p}} = \begin{pmatrix} -\boldsymbol{i}\hbar\boldsymbol{\nabla} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & -\boldsymbol{i}\hbar\boldsymbol{\nabla} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & -\boldsymbol{i}\hbar\boldsymbol{\nabla} \end{pmatrix}$

The expectation value of the momentum operator ($\hat{\boldsymbol{p}}$) is calculated as $\langle\Psi|\hat{\boldsymbol{p}}|\Psi\rangle$ under eigenstate

The matrix form of the expectation value of the momentum operator

$$\langle\boldsymbol{\phi}|\hat{\boldsymbol{p}}|\boldsymbol{\phi}\rangle = \begin{pmatrix} c_1\langle b_{11}| & \cdots & c_1\langle b_{n1}| \\ \vdots & \cdots & \vdots \\ c_1\langle b_{1n}| & \cdots & c_n\langle b_{nn}| \end{pmatrix}\begin{pmatrix} -\boldsymbol{i}\hbar\boldsymbol{\nabla} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & -\boldsymbol{i}\hbar\boldsymbol{\nabla} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & -\boldsymbol{i}\hbar\boldsymbol{\nabla} \end{pmatrix}\begin{pmatrix} c_1|b_{11}\rangle & \cdots & c_n|b_{1n}\rangle \\ \vdots & \cdots & \vdots \\ c_1|b_{n1}\rangle & \cdots & c_n|b_{nn}\rangle \end{pmatrix}$$

IV. What is the expectation value of operator?

Notes about the operator and the expectation value of operator

♦ **The expectation value of any operator ($\hat{A}$) is the measurement value of this physical observable in reality, this expectation value must be a real value; rather than the complex one** $\hat{A}|\phi\rangle = a|\phi\rangle$

♦ **In quantum mechanics, the operator ($\hat{A}$) must be Hermitian as** $\hat{A} = (\hat{A})^{\dagger}$

$$\langle\phi|\hat{A}|\phi\rangle = a \xrightarrow{\text{conjugate transpose}} (\langle\phi|\hat{A}|\phi\rangle)^{\dagger} = (|\phi\rangle)^{\dagger}(\hat{A})^{\dagger}(\langle\phi|)^{\dagger} = \left\langle\phi\middle|(\hat{A})^{\dagger}\middle|\phi\right\rangle = a^{\dagger}$$

The expectation value is real $a = a^{\dagger}$ (with $a^{\dagger} = a$)

$$\langle\phi|\hat{A}|\phi\rangle = \left\langle\phi\middle|(\hat{A})^{\dagger}\middle|\phi\right\rangle$$

$$\hat{A} = (\hat{A})^{\dagger}$$

V. What is density operator?

$$\Psi = \sum_i c_i |b_i\rangle \xleftrightarrow{\text{Conjugate Transpose}} \Psi^\dagger = \sum_j c_j \langle b_j|$$

Density operator $\hat{\rho} = \Psi\Psi^+ = \sum_i \sum_j c_i c_j |b_i\rangle \langle b_j|$

Matrix Form

$$\hat{\rho} = \begin{pmatrix} c_1|b_{11}\rangle & \cdots & c_n|b_{1n}\rangle \\ \vdots & \cdots & \vdots \\ c_1|b_{n1}\rangle & \cdots & c_n|b_{nn}\rangle \end{pmatrix} \begin{pmatrix} (c_1|b_{11}\rangle)^\dagger & \cdots & (c_1|b_{n1}\rangle)^\dagger \\ \vdots & \cdots & \vdots \\ (c_n|b_{1n}\rangle)^\dagger & \cdots & (c_n|b_{nn}\rangle)^\dagger \end{pmatrix}$$

$$= \begin{pmatrix} c_1|b_{11}\rangle & \cdots & c_n|b_{1n}\rangle \\ \vdots & \cdots & \vdots \\ c_1|b_{n1}\rangle & \cdots & c_n|b_{nn}\rangle \end{pmatrix} \begin{pmatrix} c_1\langle b_{11}| & \cdots & c_1\langle b_{n1}| \\ \vdots & \cdots & \vdots \\ c_1\langle b_{1n}| & \cdots & c_n\langle b_{nn}| \end{pmatrix}$$

V. What is density operator?

♦ **Note about the eigenstate** $\sum_i |\phi_i\rangle\langle\phi_i| = 1$

$$\Psi = \sum_i c_i|\phi_i\rangle \longrightarrow c_i = \sum_i \langle\phi_i|\Psi$$

$$\Psi = \sum_i \langle\phi_i|\Psi|\phi_i\rangle$$

number; not matrix

$$\Psi = \sum_i |\phi_i\rangle\langle\phi_i|\Psi$$

$$\sum_i |\phi_i\rangle\langle\phi_i| = 1$$

$$\Psi = \sum_i \langle b_i|\Psi|b_i\rangle$$

$$Tr(\hat{\rho}) = \sum_{ii} \Psi\Psi^{+} = 1$$

$$\Psi = \sum_i c_i|b_i\rangle$$

$$\hat{\rho} = \begin{pmatrix} a_{11} & \cdots & \cdots \\ \vdots & \cdots & \vdots \\ \cdots & \cdots & a_{nn} \end{pmatrix}$$

$$Tr(\hat{\rho}) = a_{11} + \cdots + a_{nn} = 1$$

$$\longrightarrow c_i = \langle b_i|\Psi$$

$$\longrightarrow c_i = (\langle b_i|)^{\dagger}\Psi^{\dagger} = |b_i\rangle\Psi^{\dagger}$$

$$\Psi^{\dagger} = \sum_i c_i\langle b_i|$$

V. What is density operator?

Why density operator?

$$\langle \hat{A} \rangle = Tr(\hat{A}\hat{\rho}) = Tr(\hat{\rho}\hat{A})$$ **very useful**

$$\langle \hat{A} \rangle = \langle \Psi | \hat{A} | \Psi \rangle = \langle \Psi | 1 \times \hat{A} | \Psi \rangle = \langle \Psi | \left(\sum_i |\boldsymbol{\phi}_i\rangle\langle\boldsymbol{\phi}_i| \right) \times \hat{A} | \Psi \rangle$$

$$= \sum_i \langle \Psi | \boldsymbol{\phi}_i \rangle \langle \boldsymbol{\phi}_i | \hat{A} | \Psi \rangle$$

number; not matrix

$$= \sum_i \langle \boldsymbol{\phi}_i | \hat{A} | \Psi \rangle \langle \Psi | \boldsymbol{\phi}_i \rangle$$

$$= \sum_i \langle \boldsymbol{\phi}_i | \hat{A}\hat{\rho} | \boldsymbol{\phi}_i \rangle$$

$$= Tr(\hat{A}\hat{\rho})$$



That's all for this session

Thank you for your attention and great participation